



South African traditional herbal formulation inhibits α -glucosidase, DPP-IV and glycation activities, and modulates glucose utilisation in Chang liver cells and 3T3-L1 adipocytes

M.G. Matsabisa*, C.I. Chukwuma, S.K. Chaudhary

Department of Pharmacology, Faculty of Health Sciences, University of the Free State, Bloemfontein, South Africa

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ABSTRACT

In South Africa, herbal mixture preparations are often used as alternative treatments for various diseases including diabetes. Most of the herbal mixture preparations have not been well investigated and regulated. The aim of this study was to evaluate anti-diabetic potentials and possible mode of actions of a South African herbal formulation locally called “Mabaso”. It is prepared from three plants, namely *Sclerocarya birrea* (A. Rich.) Hochst, *Opuntia ficus-indica* (L.) Mill. and *Solanum pimpinellifolium* L. The phenolic and flavonoid contents, α -glucosidase, DPP-IV and glycation inhibitory activities as well as glucose utilisation and adipocyte differentiation modulatory activities of herbal mixture was evaluated for the first time. The HPLC fingerprints of potent extracts were also determined. The DCM:MeOH extract showed the highest phenolic content. Same extract exhibited the most potent α -glucosidase, DPP-IV (73.51% at 50 μ g/ml) and glycation (62.5% at 5 μ g/ml) inhibitory activities. DCM:MeOH and MeOH extracts appreciably enhanced glucose utilisation in 3T3-L1 adipocytes and Chang liver cells respectively. Herbal mixture extracts enhanced ($p < .05$) adipocyte differentiation in 3T3-L1 adipocytes. Data suggest that the studied herbal formulation may potentiate anti-hyperglycemic effects via inhibiting α -glucosidase, DPP-IV and glycation activities as well as enhancing glucose utilisation and/or adipocyte differentiation. Thus, this herbal preparation may be an interesting candidate for further investigation on their ameliorative potentials against diabetes and its complications using appropriate animal models.

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1. Introduction

The morbidity and mortality outcomes of diabetes are prevalently increasing (IDF, 2017). Type 2 diabetes (T2D) accounts for more than 90% of total diabetes cases worldwide and currently one of the major threats to global public health (IDF, 2017). T2D is a progressive disease that is linked to several metabolic defects and organ malfunctions or damage (Pratley and Weyer 2001). Insulin resistance and impairment of pancreatic beta cell function are the most prominent pathogenic features of T2D, which progressively results in defective insulin secretion and persistent hyperglycemia (Pratley and Weyer 2001; Kahn 2003). Insulin resistance in peripheral tissues, such as adipose and muscle tissues down regulates the signalling of cellular glucose uptake, which contributes to hyperglycemia in T2D (Pratley and Weyer 2001; Kahn

2003). Also, insulin resistance causes dyslipidemia and obesity because of poor lipid metabolism and deranged adipocyte differentiation (Guilherme et al., 2008), which increases the risk of T2D and its complication. T2D complications are partly mediated by oxidative stress, due to elevated free radicals and reactive oxygen species (ROS) production. Persistent hyperglycemia promotes glucose oxidation, protein glycosylation and lipid peroxidation, which increases the physiological levels free radicals and ROS (Pourghassem-Gargari et al., 2011). Excessive levels of free radicals and ROS cause oxidative damage to biological molecules and cells (Pourghassem-Gargari et al., 2011), thus aggravating T2D complications.

There is a growing interest in the use of medicinal plants for the treatment and management of diabetes, oxidative stress and related diseases, because plants are endowed with rich therapeutic phytochemicals. Herbal medicine is also relatively cheaper, poses lesser side effects and more accessible compared to synthetic drugs (Yuan et al., 2017).

In folk medicine, the synergistic effect of plants is well known. Most herbal medicines contain different plants or plant parts, which can synergistically enhance efficacy or reduce toxicity when compared to individual isolated compounds (Yuan et al., 2017). In South Africa, most traditional health practitioners prepare their medications as herbal cocktails of two or more plants, perhaps due to their belief in

Abbreviations: AGEs, advanced glycation end products AGEs; DCM, dichloromethane extract; DCM:MeOH, 1:1 dichloromethane and methanol extract; DPP-IV, dipeptidyl peptidase IV; GIP, gastric inhibitory polypeptide or glucose-dependent insulinotropic polypeptide; GLP-1, glucagon-like peptide-1; MeOH, methanol extract; PPAR γ , peroxisome proliferator-activated receptor gamma.

* Corresponding author at: Department of Pharmacology, Faculty of Health Sciences, University of the Free State, PO Box 339, Bloemfontein 9300, South Africa.

E-mail address: MatsabisaMG@ufs.ac.za (M.G. Matsabisa).

phytomedicinal synergy. In this study we focused on a medicinal herbal formulation locally called “Mabaso”. It is made from three plants found in South Africa, namely *Sclerocarya birrea* (A. Rich.) Hochst, *Opuntia ficus-indica* (L.) Mill. and *Solanum pimpinellifolium* L.

Sclerocarya birrea, commonly known as “Marula” is a medium-sized dioecious tree of the Anacardiaceae family (Prinsloo and Street 2013). It is indigenous to Southern Africa, where its fruits are consumed as food for man, baboons and elephants. The leaves and stem-bark are traditionally used to treat diarrhoea, dysentery, proctitis, stomach ailments, ulcers, inflammation, arthritis, hypertension, skin diseases, fever, malaria and diabetes mellitus (Ojewole et al., 2010). Its stem bark has been reported to possess pharmacological properties associated with the treatment of several ailments, including diabetes (Ojewole et al., 2010; Prinsloo and Street 2013). *Opuntia ficus-indica*, also called “prickly pear” is a cactus species of the family Cactaceae. Juice from the cladodes is used in African traditional medicine to treat diabetes and wound infections (Werner et al., 2013). Different parts of this plant, particularly the juicy or succulent cladodes, have been a source of human nutrients (Hernández-Urbiola et al., 2011; Osuna-Martínez et al., 2014) and pharmacological agents with anti-inflammatory, antioxidant, hypoglycemic, antimicrobial, anticancer, antilipidemic, neuroprotective properties (Zou et al., 2005; El-Mostafa et al., 2014; Osuna-Martínez et al., 2014). *Solanum pimpinellifolium*, also known as “currant tomato” is a wild species of tomato belonging to the Solanaceae family. Traditionally, the sliced fruit is used as a first aid for treating skin burns (Allardice 1993), while root decoction is ingested as a remedy for tooth ache (Duke and Ayensu, 1985). Although there are no documented reports on the diabetes-related medicinal properties of *S. pimpinellifolium*, it is rich in lycopene, a bioactive agent that has shown protective effect against oxidative stress, cardiovascular disease, hypertension, atherosclerosis, skin damage, and certain cancers (Cámara et al., 2013).

The studied herbal mixture was formulated by a South African traditional health practitioner, who has been using it to treat diabetic patients in South Africa for almost a decade. The stem bark of marula, slimy juice from succulent prickly pear pads and dried fruits of currant tomato are mixed in given proportions to make a cocktail decoction for treating diabetes. However, this medicinal claim of the herbal formulation has not been scientifically investigated and/or validated in a scientifically controlled setting. Such scientific investigation and/or validation may potentially promote clinical and economic relevance of this herbal formulation and ensure its proper regulation. Therefore, the present study, for the first time, investigated the antihyperglycemic potentials and possible mode actions of this traditional anti-diabetic herbal formulation using various experimental models. A consent was obtained from the traditional health practitioner, who formulated the herbal mixture to conduct this study under a signed research agreement.

2. Materials and methods

2.1. Reagents and chemicals

Dinitrosalicylic acid, p-nitro-phenyl- α -D-glucopyranoside (p-NPG), rat intestinal acetone powder, gly-pro-p-nitroanilide, human dipeptidyl peptidase IV (DPP-IV), acarbose, aminoguanide, insulin, rosiglitazone, metformin, Folin–Ciocalteu solution and HPLC assay solvents were purchased from Sigma Aldrich, South Africa. Chang liver cells, 3T3-L1 adipocytes were purchased from ATCC.

2.2. Plant collection and extraction

The stem bark of Marula (*S. birrea*) was collected in Bronkhorstspuit in the Limpopo province of South Africa (Voucher

specimen no. GLOW 711985; deposited with the Botanical Gardens in Nelspruit, Mpumalanga, South Africa). The leaf pads of prickly pear (*O. ficus-indica*) was collected in Botshabelo, Free State Province of South Africa, while whole plant of currant tomato (*S. pimpinellifolium*) was collected in Cape town, Western Cape province of South Africa (specimen voucher numbers: BLFU MGM003 and BLFU MGM005, respectively; deposited and authenticated at the Geo Potts herbarium, University of the Free State, Bloemfontein, South Africa). The stem bark of marula and the fruits of currant tomato were air-dried at ambient conditions, while the slimy juice from the succulent prickly pear pads were mulched-frozen immediately and freeze-dried. The dried plant parts were pulverised into fine powder and mixed in proportions consistent with the herbal formulation used by the traditional health practitioner. The plant powder mixture (100 g in a 1:5 mass:solvent ratio) was extracted using hexane, DCM, DCM:MeOH (1:1, v/v) and MeOH sequentially. Extraction was done for 24 h, at ambient room temperature with mild agitation of 100 rpm. After extraction, each solvent was filtered and concentrated using an R – 215 rotary evaporator (Buchi, Switzerland). Concentrated extracts were air-dried and stored at 4 °C. The crude extracts were reconstituted in methanol or DMSO to prepare various concentrations used for phytochemical characterisation as well as for in vitro and cell culture assays.

2.3. High-performance liquid chromatography

HPLC-diode array detection analysis was performed using an Agilent 1100 series (Agilent, Waldbronn, Germany) instrument equipped with photo diode array, autosampler, column thermostat and degasser. A Phenomenex: Luna 5 μ m C₁₈ (2) (150 $\times$ 4.6 mm; 5 μ m particle size) column was used as the stationary phase. Water containing 0.01% of formic acid (A) and acetonitrile (B) served as mobile phases at a flow rate of 1 ml/min. Gradient elution was applied as follows: Initial ratio 95% A: 5% B, keeping for 10 min, changed to 90% A: 10% B in 10 min, changed to 70% A: 30% B in 10 min, to 50% A: 50% B in 10 min, maintaining for 0.5 min and back to initial ratio in 0.5 min. Temperature was set to 30°C. The injection volume was 20.0 μ L and chromatograms were recorded at 254 nm.

2.4. Total phenolic and flavonoid content estimation

Total phenolics and flavonoids were estimated according to a method reported previously (Lee et al., 2015). Content was expressed in milligram gallic acid (for phenolic content) or quercetin (for flavonoid content) equivalent per gram of extract.

2.5. In vitro assays

2.5.1. Alpha glucosidase inhibition

Alpha glucosidase inhibition was measured spectrophotometrically using the synthetic substrate p-nitrophenyl α -D-glucopyranoside in 0.01 M phosphate buffer pH 6.9 system according to a method previously reported (Chukwuma and Islam 2015). Alpha glucosidase enzyme solution was prepared from commercially sourced rat intestinal acetone powder. The samples and acarbose (positive control) were screened at 125 μ g/ml in three replicates.

2.5.2. DPP-IV inhibition

Inhibitory activity was determined using the chromogenic substrate Gly-Pro-pNA and recombinant human DPP-IV according to the method of a previous report (Matheeußen et al., 2012). The final concentration of extract in reaction mixture was 50 μ g/ml.

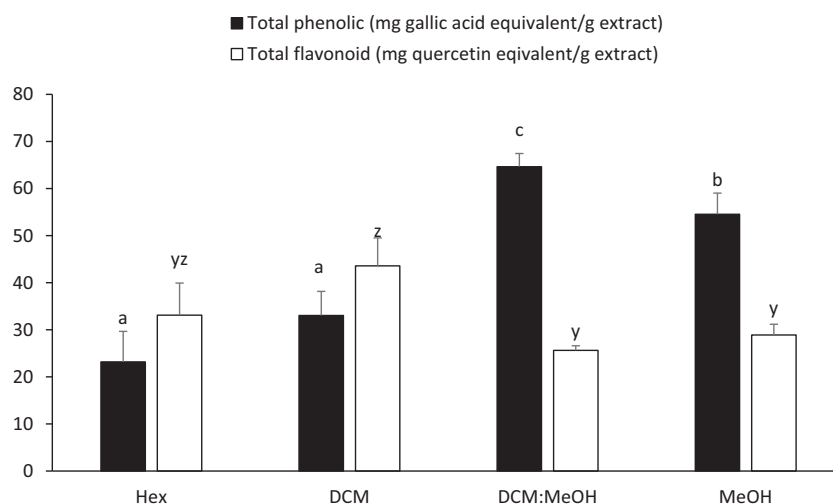


Fig. 1. Total phenolics and flavonoids contents of different extracts of herbal mixture. Data presented as mean $\pm$ SD of three replicates (“abc” and “yz” data labels mean values are significantly different from each other when comparing extracts total phenolic and flavonoid respectively; $p < .05$; Tukey’s HSD multiple range post-hoc test, IBM, SPSS, version 23).

2.5.3. Antiglycation activity

Antiglycation was evaluated as the inhibition of advanced glycation end products (AGEs) using bovine serum albumin (BSA) as substrate in the presence of glucose according to the method of a previous study (Grzegorzczak-Karolak et al., 2016). AGEs formed during the 24-h incubation at 60 °C were estimated using their fluorescent properties. Each extract (5 and 50 μ g/ml) or 10 mM aminoguanidine (positive control) was tested in triplicate. Percentage inhibition was calculated as follows:

$$\% \text{Inhibition} = (\text{Fc} - \text{Fs}) / \text{Fc} \times 100$$

where, “Fc” and “Fs” were fluorescence of the control (without inhibitor) and test samples (extract or inhibitor) respectively.

2.6. Cell culture assays

2.6.1. Glucose utilisation study in 3T3-L1 adipocytes and Chang liver cells

Assays were performed according to a previously reported method (Oyedemi et al., 2013). Extracts were screened at a concentration of 50 μ g/mL. A 1 μ M insulin and 1 μ M metformin were used as positive control for glucose utilisation in Chang liver cells and 3T3-L1 adipocytes respectively. All assays were performed in six replicates. Glucose utilisation/uptake of control and test were expressed as a percentage of the negative control, where control was 100% of negative control.

2.6.2. 3T3-L1 adipocyte differentiation study

Triglyceride accumulation (a marker for adipocyte differentiation) was measured by Oil red-O staining according to a method reported previously (Hasan et al., 2017). Extracts were screened at a concentration of 25 μ g/ml. A 0.8 μ M rosiglitazone was used as positive control. All assays were performed in six replicates. Adipocyte differentiation of control and test were expressed as a percentage of the negative control, where control was given as 100% of negative control. Extracts showing modulatory effects on adipocyte differentiation were subjected to a dose-dependent (25, 50 and 100 μ g/ml) evaluation.

2.7. Statistical analysis

Results were analysed using one-way analysis of variance (ANOVA) and Tukey’s multiple range post hoc test (IBM, SPSS, version 23) and p values less than 0.05 were considered significant, when comparing groups.

3. Results

3.1. Yield and phytochemical chemical composition

Extracts yields were 0.09%, 0.23%, 13.74% and 1.3% for hexane, DCM, DCM:MeOH and MeOH respectively. Phenolic content was higher ($p < .05$) in DCM:MeOH and MeOH compared to hexane and DCM extracts (Fig. 1). The HPLC finger prints of DCM:MeOH and MeOH extracts shared some peak similarities at retention times 4 min (4.518 with % peak area 17.140 and 4.566 min with peak value 19.001, respectively), 8 min (8.613 min with % peak area 31.912 and 8.666 min with % peak area 34.711, respectively) and 19 min (19.55 min with % peak area 27.399 and 19.567 min with % peak area 24.314, respectively) (Fig. 2a and b).

3.2. Inhibitory activity of plant extracts on α -glucosidase and DPP-IV

Results are presented in Table 1. The DCM:MeOH extract consistently showed the most potent ($p < .05$) inhibitory activity on α -glucosidase (43.77%) and DPP-IV (73.51%), although its α -glucosidase inhibitory was not comparable ($p < .05$) to that of acarbose (positive control) (56%) (Table 1).

3.3. Anti-glycation effect of extracts

The ability of the extracts to inhibit the glucose-induced glycation of BSA and subsequent AGEs formation is presented in Fig. 3. While the hexane extract favoured AGEs formation, DCM, DCM:MeOH and MeOH inhibited the process at 5 and 50 μ g/ml concentrations. At 5 μ g/ml, anti-glycation effect of DCM:MeOH extract was higher than that of DCM ($p < .05$) and MeOH extracts, when no significant difference was observed at 50 μ g/ml (Fig. 3). The anti-glycation effect of 10 mM Aminoguanidine (standard drug) was more potent ($p < .05$) than that of all tested extracts.

3.4. Effect of extracts on glucose utilisation in Chang liver cells and 3T3-L1 adipocytes

The glucose utilisation results are presented in Table 2. Although not as effective as 1 μ M Insulin, the MeOH extracts exhibited appreciable modulatory effect on glucose utilisation in 3T3-L1 adipocytes, which was significantly more effective ($p < .05$) than that of other extracts. In Chang liver cells, the glucose uptake modulatory effects of the DCM:MeOH extract was more potent ($p < .05$) than that of DCM

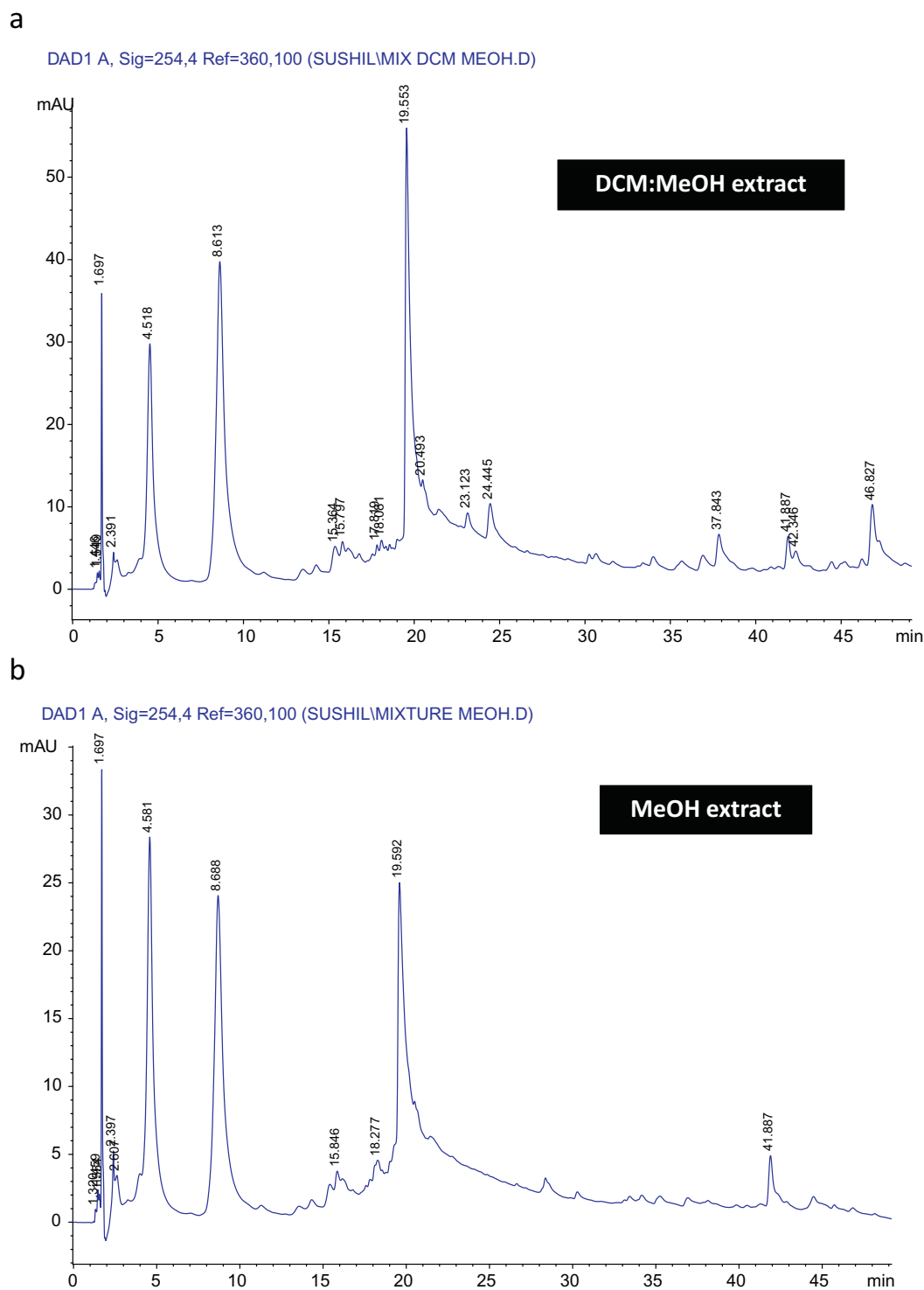


Fig. 2. HPLC chromatogram of DCM:MeOH and MeOH extracts of herbal mixture.

and MeOH extracts and 1 μ M Metformin, when the effect of DCM and MeOH did not significantly differ from that of 1 μ M Metformin (Table 2).

3.5. Effect of extracts on 3T3-L1 adipocytes differentiation

Fig. 4 shows the effects of plant extracts on adipocyte differentiation in 3T3-L1 cells. All extracts caused an increase in adipocyte

differentiation at tested concentrations. Rosiglitazone standard drug was more active ($p < .05$) than all tested extracts.

4. Discussion

Medicinal plants have been an integral part of traditional medicine for a long time. African herbal medicines are mostly prepared as cocktail mixtures of different plants or plant parts to possibly

Table 1
Percentage inhibition of α -glucosidase and DPP-IV.

Extract or treatment	Inhibition (%)	
	α -glucosidase [#]	DPP-IV
Acarbose	56.07 $\pm$ 2.60 ^d	ND
Hexane	7.65 $\pm$ 2.94 ^a	0
DCM	4.07 $\pm$ 3.30 ^a	0
DCM:MeOH	43.77 $\pm$ 1.66 ^c	73.51
MeOH	16.69 $\pm$ 5.45 ^b	0

[#] Data presented as mean $\pm$ SD of six replicates ("abcd" superscripts mean values are significantly different from each other when comparing different treatment groups; $p < .05$; Tukey's HSD multiple range post-hoc test, IBM, SPSS, version 23). ND means not determined.

enhance the therapeutic effects of medicinal plants. The present study focused on a South African herbal mixture made from three different plants. This herbal mixture was formulated and is being used by a traditional health practitioner in treating diabetes. The aim was to investigate the possible mode of actions underlying the anti-diabetic potentials of this herbal formulation.

We estimated the α -glucosidase and dipeptidyl peptidase-IV (DPP-IV) inhibitory potentials of the herbal mixture. Alpha glucosidase inhibitors ameliorate hyperglycemia in diabetics by impairing the action of carbohydrate digesting enzymes (α -glucosidase and α -amylase) (Patel et al., 2012). Dipeptidyl peptidase-IV (DPP-IV) inhibitors suppress the proteolytic action of DPP-IV on gastric inhibitory polypeptide (GIP) and glucagon-like peptide-1 (GLP-1). GIP and GLP-1 are incretin hormones that promote glucose homeostasis by stimulating pancreatic insulin secretion, β -cell growth and differentiation and inhibiting glucagon secretion (Ahren, 2005). Thus, preserving their functionality by inhibiting DPP-IV-induced inactivation is crucial for modulating glucose homeostasis.

In the present study, the DCM:MeOH extract of herbal mixture, showed promising inhibitory activity on α -glucosidase and DPP-IV (Table 1). The DPP-IV inhibitory effect of this extract is worthy of note, with about 73.51% inhibition at 50 μ g/ml. Thus, may potentiate anti-hyperglycemic effect via modulating incretin hormone-associated glucose homeostasis (Ahren 2005). The high phenolic content of the DCM:MeOH extract (Fig. 1) suggest that it may possess phenolics associated with α -glucosidase (Tan et al., 2013) and DPP-IV (Domínguez et al., 2017) inhibition.

Table 2

Percentage glucose utilisation by 3T3-L1 adipocytes and Chang liver cells of different solvent extracts of herbal mixture.

Extract or treatment	Glucose utilisation (%) [#]	
	3T3-L1 adipocytes	Chang liver cells
1 μ M Insulin	156.24 $\pm$ 11.31 ^d	ND
1 μ M Metformin	ND	122.41 $\pm$ 6.55 ^y
Hexane	98.63 $\pm$ 3.15 ^a	98.38 $\pm$ 4.93 ^x
DCM	109.06 $\pm$ 0.74 ^b	128.34 $\pm$ 7.41 ^y
DCM:MeOH	92.59 $\pm$ 13.19 ^a	156.27 $\pm$ 11.68 ^z
MeOH	136.43 $\pm$ 9.42 ^c	131.09 $\pm$ 7.93 ^y

[#] Glucose utilisation expressed as % of negative control, when negative was given as 100% (Differences in cell numbers were corrected for by determining viable cell numbers with the MTT assay and normalising glucose utilisation results). Data presented as mean $\pm$ SD of six replicates ("abcd" and "xyz" superscripts mean values are significantly different from each other when comparing different treatment in 3T3-L1 adipocytes and Chang liver cells respectively; $p < .05$; Tukey's HSD multiple range post-hoc test, IBM, SPSS, version 23). ND means not determined.

Protein and lipid glycation and subsequent formation AGEs are major culprits of diabetic complications due to physiological oxidative damage (Cooper 2004). The role of polyphenols as reactive carbonyl scavengers and glycation inhibitors has been well documented (Yeh et al., 2017). In the present study, inhibition of AGEs formation was used as an index to measure the anti-glycation potential of herbal mixture. We observed that all of the tested herbal mixture extracts, excluding the hexane extract, inhibited AGEs formation (Fig. 3). However, the most potent anti-glycation activity was demonstrated by the DCM:MeOH extract at 5 μ g/ml. The DCM:MeOH extract concomitantly showed the highest polyphenol content (Fig. 1), which suggest the presence of phenolics that may ameliorate oxidative stress-related complications of diabetes.

Impaired insulin-mediated hepatic glucose utilisation and storage is a known metabolic derangement in diabetic patients (especially type 2) that contributes to hyperglycaemia in patients (Chukwuma et al., 2016, 2017). Insulin resistance in type 2 diabetic subjects impairs the uptake and utilisation of glucose by cells of peripheral tissue including muscle cells and adipocytes, resulting in poor systemic glucose homeostasis (Chukwuma et al., 2016, 2017). Hence, we investigated the effects of the herbal mixture on glucose uptake and utilisation in hepatocytes and adipocytes as possible antidiabetic mode of action. We observed that the DCM:MeOH and MeOH extracts exhibited the most potent glucose utilisation modulatory activities in Chang liver cells and 3T3-L1

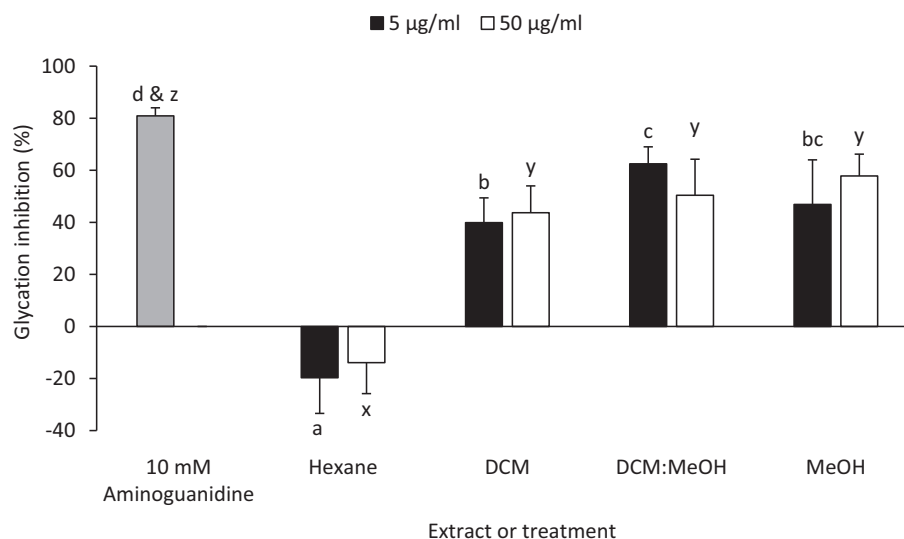


Fig. 3. Percentage glycation inhibition of different solvent extracts of herbal mixture. Data presented as mean $\pm$ SD of three replicates ("abc" and "xyz" data labels mean values are significantly different from each other when comparing different treatment for 5 and 50 μ g/ml respectively; $p < .05$; Tukey's HSD multiple range post-hoc test, IBM, SPSS, version 23).

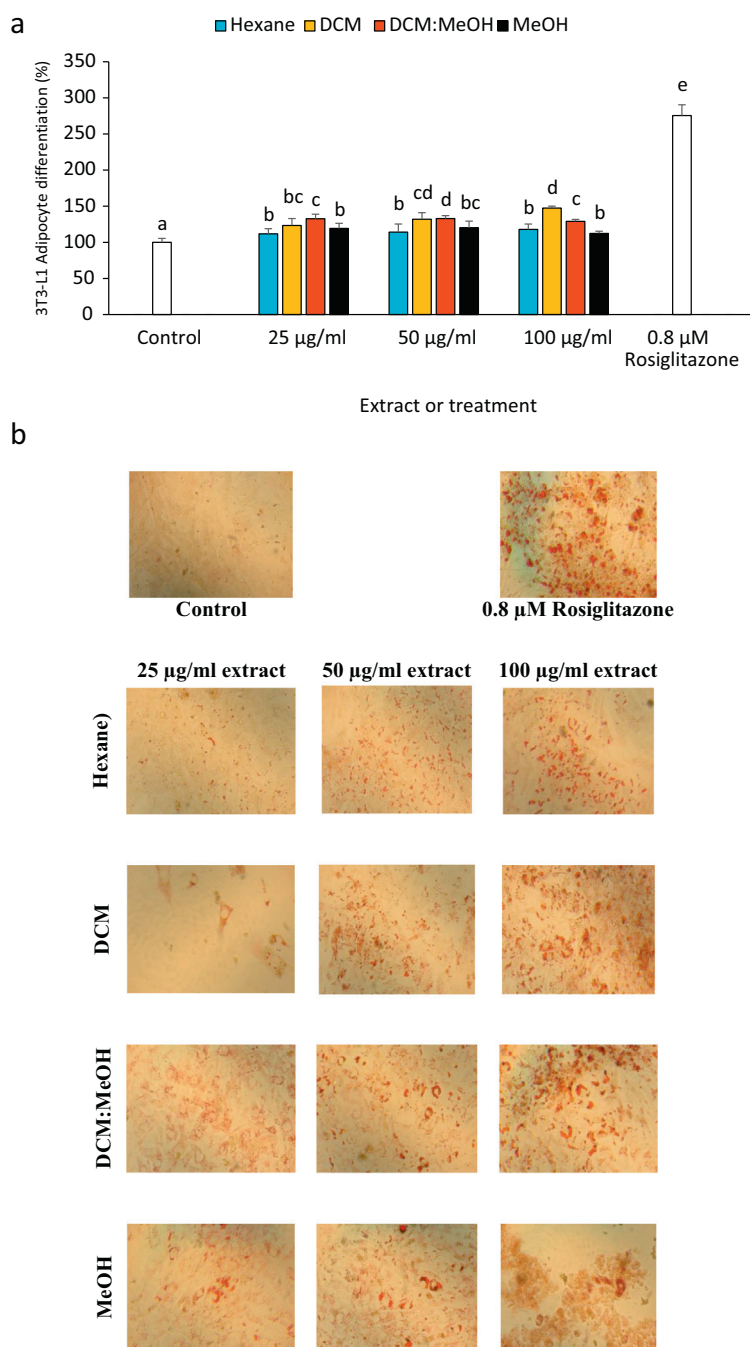


Fig. 4. Graph (a) and microscopic images (b) showing 3T3-L1 Adipocyte differentiation (%) of herbal mixture extracts. Values expressed as % of negative control, when control was given as 100%. Data presented as mean $\pm$ SD of six replicates ("abcd" data labels mean values are significantly different from each other when comparing among extracts at a given concentration and with normal and positive controls; $p < .05$; Tukey's HSD multiple range post-hoc test, IBM, SPSS, version 23).

adipocytes, respectively (Table 2). The modulatory effects of these extracts in Chang liver cells (Table 2) was more potent than the effect of 1 μ M Metformin.

The notable modulatory effects of the DCM:MeOH and MeOH extracts on cellular glucose utilisation (Table 2) may be linked to their phenolic content (Fig. 1), since the role of phenolics as glycemic control agents and modulators of hepatic glucose uptake has been well documented (Vinayagam et al., 2016; Ho et al., 2017). Also, HPLC finger prints revealed that the DCM:MeOH and MeOH shared some peak similarities (Fig. 2a and b), which indicate the presence of similar phytochemicals that may influence their potent activities on cellular glucose uptake/utilisation. However, this hypothesis is subject to further phytochemical and pharmacological studies. Data observed in this study

suggest that the studied herbal formulation may potentiate glycemic control via modulating systemic glucose uptake and utilisation.

Furthermore, insulin sensitivity and glucose homeostasis are partly regulated by lipid metabolism, particularly in adipocytes. Thus, the peroxisome proliferator-activated receptor gamma (PPAR γ), an essential regulator of adipocyte function is a known therapeutic target for the treatment of diabetes (Hammarstedt et al., 2005). Activation of the PPAR γ receptor promotes the differentiation of preadipocytes into insulin sensitive small adipocytes (Danforth 2000). Down-regulation of PPAR γ receptor promotes adipocyte dysfunction, which is characterised by decreased adipocyte insulin responsiveness, altered adipokine secretion and increased lipolysis, contributing to the development of whole body insulin resistance and overt diabetes (Goossens 2008). In the

present study, extracts of herbal mixture enhanced 3T3-L1 adipocytes differentiation at tested concentrations (Fig. 4a & 4b), which may be an indication of modulated insulin sensitivity in these adipocytes. The concomitant MeOH extract-induced enhancement of glucose utilisation (Table 2) and differentiation in 3T3-L1 adipocytes (Fig. 4a & 4b) further supports this correlation. This observation suggests that the studied herbal formulation may enhance insulin sensitivity via modulation of PPAR γ receptor but requires further investigation in a high-fat-diet type 2 diabetes animal model.

5. Conclusion

This study demonstrated that the studied South African traditional herbal mixture exhibited anti-diabetic potentials, at least under in vitro conditions via inhibiting α -glucosidase, DPP-IV and glycation activities as well as enhancing glucose utilisation and adipocyte differentiation in hepatocytes and/or adipocytes. The DCM:MeOH and MeOH extracts appeared to be the most potent extracts, which also shared some peak similarities in their HPLC finger prints. The studied herbal mixture can be a potential candidate for further investigation in the management of diabetes and related complications. It is also recommended that the synergistic mechanism of action of the herbal mixture be considered for further studies.

Conflict of interests

The authors declare that there are no conflicts of interest regarding the publication of this paper.

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